

Loss Functions & Risk

Lecture 13

1) Review

With linear regression we minimized ~~sum~~^{mean} squared error

$$\text{minimize } J(\theta) = \frac{1}{2m} \sum_{i=1}^m \underbrace{(h_{\theta}(x^{(i)}) - y^{(i)})^2}_{\text{loss function}}$$

cost function

A loss function in this case measures the error for a single ^{data}_(sample) point. The cost function~~s~~ measures some statistic across a set of samples.

With ^{binary} classification, every classification is either right or wrong. If there are more than two choices (classes) then a classification is either right or a match to one of the incorrect classes. We can minimize error by always picking the class with the highest probability. The probability of error thus becomes the probability that the ~~classifi~~^{true} class is any other class than the ~~the~~ class with the highest probability. When there are two classes this becomes

$$\text{minimize } p(\text{error}) = \int_{\bar{x} \in \mathbb{R}^n} \min(P(w_1 | \bar{x}), P(w_2 | \bar{x})) p(\bar{x}) d\bar{x}$$

If we consider the above a cost function, it is a cost function that error has equivalent to ~~loss~~ cost.

What if cost $\neq$ error?

Sometimes cost is higher ~~if~~ ~~we~~ depending on the type of error. This is why we sometimes bias a classifier toward precision or recall. We presented examples in the last lecture.

To deal with the case where not all ^{types} ω_i result in the same cost, we need to introduce some new concepts.

$$\lambda_{ij} \triangleq \lambda(\alpha_i | \omega_j) = \text{the loss derived from taking action } \alpha_i$$

What are actions?

Detecting cancer is a decision.

The action might be chemotherapy.

Chemotherapy introduces real health risks.

What is risk?

In this discussion, risk = expected cost.

We measure across all possible actions we can take in response to a decision.

Example: drone strike

A drone classifies targets as "friend" or "foe."

ω_1 = friend, ω_0 = foe

α_1 = spare the target (do nothing), α_0 = eliminate the target

$$\lambda_{11} = \lambda(\alpha_1 | \omega_1) = 0 \quad \leftarrow \text{spare a friend}$$

$$\lambda_{10} = \lambda(\alpha_1 | \omega_0) = 1 \quad \leftarrow \text{spare a foe}$$

$$\lambda_{01} = \lambda(\alpha_0 | \omega_1) = 100 \quad \leftarrow \text{eliminate a friend}$$

$$\lambda_{00} = \lambda(\alpha_0 | \omega_0) = 0 \quad \leftarrow \text{eliminate a foe}$$

We place a much higher loss on eliminating a friend than on sparing a foe.

The choice of costs here is arbitrary, but we must assign some finite cost on eliminating a friend or the resulting system would spare everyone.

Risk

risk = expected value of the cost function

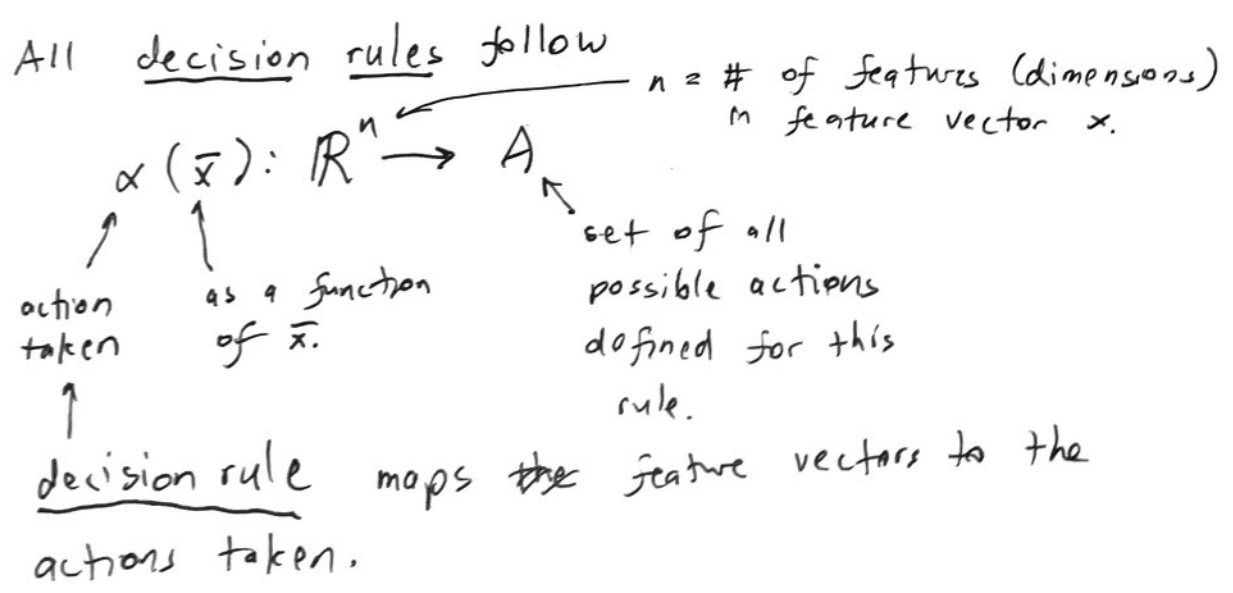
conditional risk:

$$R(\alpha_i | \bar{x}) = \sum_{j=1}^k \lambda(\alpha_i | \omega_j) P(\omega_j | \bar{x}) \quad \text{where } k = \# \text{ of classes}$$

The conditional risk is the expected loss given feature vector $\bar{x}$.

How we assign losses to each action allows us to bias the decision rule. A higher cost for an action will bias decisions away from that action.

Given the objective of minimizing risk, what decision rule should we employ?



Bayes' Risk

$$R(\alpha) = \int_{\bar{x} \in \mathbb{R}^n} R(\alpha(\bar{x}) | \bar{x}) P(\bar{x}) d\bar{x}$$

Bayes' Risk is the expected loss (risk) for a given decision rule α .

Bayes' Risk (cont.)

α^* denotes the decision rule that minimizes Bayes' risk and is called Bayes' Decision Rule.

$$R(\alpha^*) = \int_{\bar{x} \in \mathcal{R}^n} \left[\min_{i=1, \dots, M} R(\alpha_i, \bar{x}) \right] P(\bar{x}) d\bar{x}$$

of actions

~~Any~~ Any given decision α_i that increases $R(\alpha_i, \bar{x})$ the integral would increase. That isn't a proof but just ponder it a bit. Is there any way that choosing the riskier action given $\bar{x}$ could decrease the expected risk?

More formally, for any decision rule α

$$R(\alpha) = \int_{\bar{x} \in \mathcal{R}^n} R(\alpha|\bar{x}) p(\bar{x}) d\bar{x} \geq \int_{\bar{x} \in \mathcal{R}^n} \left[\min_{i=1, \dots, M} R(\alpha_i, \bar{x}) \right] P(\bar{x}) d\bar{x}$$

i.e.,

$$R(\alpha) \geq R(\alpha^*)$$

Example 1. Binary classification

$$\lambda_{ij} = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} \text{ where } \lambda_{12} > \lambda_{22} \text{ and } \lambda_{21} > \lambda_{11}$$

cost of taking α_2 when class is ω_1

use Bayes' decision rule.

decide $\begin{cases} \omega_1 & \text{if } R(\alpha_1|\bar{x}) < R(\alpha_2|\bar{x}) \\ \omega_2 & \text{otherwise.} \end{cases}$

cost of a mistake

cost of a correct action

Example 1 (cont.)

Using the definition of the conditional risk (1)

(4)

we can rewrite (3) as

$$\text{decide } \begin{cases} \omega_1 & \text{if } \frac{\lambda_{11} p(\bar{x}|\omega_1) p(\omega_1) + \lambda_{12} p(\bar{x}|\omega_2^*) p(\omega_2)}{p(\bar{x})} \\ & < \frac{\lambda_{21} p(\bar{x}|\omega_1) p(\omega_1) + \lambda_{22} p(\bar{x}|\omega_2) p(\omega_2)}{p(\bar{x})} \\ \omega_2 & \text{otherwise} \end{cases}$$

The probability $p(\bar{x})$ is the same on both sides of the inequality in (4), so it cancels out leaving

$$\text{Decide } \begin{cases} \omega_1 & \text{if } \lambda_{11} p(\bar{x}|\omega_1) p(\omega_1) + \lambda_{12} p(\bar{x}|\omega_2) p(\omega_2) \\ & < \lambda_{21} p(\bar{x}|\omega_1) p(\omega_1) + \lambda_{22} p(\bar{x}|\omega_2) p(\omega_2) \\ \omega_2 & \text{otherwise} \end{cases} \quad (5)$$

Rearranging (5) yields

$$\text{Decide } \begin{cases} \omega_1 & \text{if } (\lambda_{11} - \lambda_{21}) p(\bar{x}|\omega_1) p(\omega_1) < (\lambda_{22} - \lambda_{12}) p(\bar{x}|\omega_2) p(\omega_2) \\ \omega_2 & \text{otherwise} \end{cases} \quad (6)$$

Likelihood ratios

A "likelihood ratio" is the ratio of the probability of one observation (a sample) given it came from one class and for that same observation coming from another class

$$\left. \frac{p(\bar{x}|\omega_1)}{p(\bar{x}|\omega_2)} \right\} \text{likelihood ratio}$$

(7)

Likelihood ratios

Let's continue example 1 using likelihood ratios

From (6) and (7)

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{P(\bar{x}|\omega_1)}{P(\bar{x}|\omega_2)} > \frac{\lambda_{22} - \lambda_{12}}{\lambda_{11} - \lambda_{21}} \frac{P(\omega_2)}{P(\omega_1)} \\ \omega_2 & \text{otherwise} \end{cases} \quad (8)$$

comparison
flipped because

$$\lambda_{22} < \lambda_{12}$$

$$\text{thus } \lambda_{11} - \lambda_{21} < 0$$

we divided both sides by
a negative number.

Likelihood Ratios with Priors

move the prior probability ratio from the right hand side of
(8) to the left hand side

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{P(\bar{x}|\omega_1)}{P(\bar{x}|\omega_2)} \frac{P(\omega_1)}{P(\omega_2)} > \frac{\lambda_{22} - \lambda_{12}}{\lambda_{11} - \lambda_{21}} \\ \omega_2 & \text{otherwise} \end{cases} \quad (9)$$

Likelihood Ratios w/ Priors (cont.)

for binary classification

How would we define a loss function that minimizes error (i.e., maximizes accuracy)?

- 1) we would expect zero loss for obtaining the correct answer.

$$\lambda_{11} = \lambda_{22} = 0$$

- 2) we would expect the same loss for making either incorrect answer.

$$\lambda_{12} = \lambda_{21} = 1$$

With this ~~loss~~ assignment the optimal classifier uses the decision rule

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{P(\bar{x} | \omega_1) P(\omega_1)}{P(\bar{x} | \omega_2) P(\omega_2)} > \frac{0-1}{0-1} \\ \omega_2 & \text{otherwise} \end{cases}$$

which becomes

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{P(\bar{x} | \omega_1) P(\omega_1)}{P(\bar{x} | \omega_2) P(\omega_2)} > 1 \\ \omega_2 & \text{otherwise} \end{cases} \quad (10)$$

Likelihood Ratios w/ Priors (cont.)

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In practice, we should avoid ratios as a decision rule for at least the following reasons

1) Numerical stability

Likelihoods are often products of many probabilities. Probabilities ~~are~~ ^{are} always ≤ 1 . ~~when~~ Thus a product of probabilities will always shrink. A product of many probabilities that are already individually small can result in numerical underflow.

2) Difficult to interpret ~~due to large dynamic range~~
Values with a wide range in the magnitude are difficult to visualize. Often times when dealing with values that may have ~~very~~ wildly different order of magnitude, we use log plots.

A common alternate way of dealing with likelihoods is to use log of the likelihoods.

Additional benefits of log likelihoods

1) Easier math

In ML, likelihoods are products of probabilities, a log of a product converts a product into a sum.

representation 1
 ab

representation 2

$$\log(ab) = \log a + \log b$$

1) Easier math (cont.)

Using ~~log~~ sum of logs simplifies taking derivatives and thus simplifies gradient descent. Calculating the gradient of the log likelihood is easier.

Log Likelihood Ratios

$$\log \left(\frac{P(\bar{x}|w_0) P(w_0)}{P(\bar{x}|w_1) P(w_1)} \right) = \log p(\bar{x}|w_0) + \log p(w_0) - \log p(\bar{x}|w_1) - \log p(w_1)$$

The decision ~~function~~ rule, ^{when minimizing error} becomes

$$\begin{aligned} &\text{declare } w_0 \text{ if } \log p(\bar{x}|w_0) + \log p(w_0) - \log p(\bar{x}|w_1) - \log p(w_1) > 0 \\ &w_1 \text{ otherwise} \end{aligned} \quad (11)$$

When we have a loss function λ_{ij} such that

$$\lambda = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} \text{ where } \left. \begin{array}{l} \lambda_{12} > \lambda_{22} \\ \lambda_{21} > \lambda_{11} \end{array} \right\} \begin{array}{l} \text{loss of taking} \\ \text{the wrong action} \\ \text{is greater than the} \\ \text{loss of the correct} \\ \text{action} \end{array}$$

the ~~log~~ likelihood ratio w/ priors was given by (9).

(9) can be restated as a log likelihood ratio w/ priors as

$$\text{declare } \begin{cases} w_1 & \text{if } \log \left(\frac{P(\bar{x}|w_1)}{P(\bar{x}|w_2)} \cdot \frac{P(w_1)}{P(w_2)} \right) > \log \left(\frac{\lambda_{22} - \lambda_{12}}{\lambda_{11} - \lambda_{21}} \right) \\ w_2 & \text{otherwise} \end{cases} \quad (12)$$

to make the denominator and numerator both ~~no~~ positive, multiply the cost ratio by $\frac{1}{\lambda_{21} - \lambda_{11}}$ to get $\frac{\lambda_{12} - \lambda_{22}}{\lambda_{21} - \lambda_{11}}$

Rearranging (12) yields

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$$\text{declare } \begin{cases} \omega_1 & \text{if } \log P(\bar{x}/\omega_1) + \log P(\omega_1) - \log P(\bar{x}/\omega_2) - \log P(\omega_2) \\ & > \log(\lambda_{12} - \lambda_{22}) - \log(\lambda_{21} - \lambda_{11}) \\ \omega_2 & \text{otherwise} \end{cases}$$

Moving all logs to the lefthand side of the inequality yields

$$\text{declare } \begin{cases} \omega_1 & \text{if } \log P(\bar{x}/\omega_1) + \log P(\omega_1) - \log P(\bar{x}/\omega_2) \\ & - \log P(\omega_2) - \log(\lambda_{12} - \lambda_{22}) + \log(\lambda_{21} - \lambda_{11}) > 0 \\ \omega_2 & \text{otherwise} \end{cases} \quad (13)$$

(13) expresses the decision rule for a binary classifier that minimizes risk for the cost function λ .

Increasing the loss incurred by a mistake means increasing

λ_{12} and/or λ_{21} . In (13), increasing λ_{12} also increases $\log(\lambda_{12} - \lambda_{22})$ which requires larger values for $\log P(\bar{x}/\omega_1) + \log P(\omega_1)$ to overcome the loss for taking the action for deciding ω_1 .

Increasing λ_{12} biases the classifier towards deciding ω_2 .

If ω_2 means "person" and ω_1 means not person, the system would be biased toward identifying a person.

